Paper I: The ARC Principle

FULL TITLE The ARC Principle: Formalisation and Preliminary Validation of Recursive Capability Scaling

VERSION v1.1

FIRST PUBLISHED 17 January 2026

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This paper formalises and preliminarily tests the ARC Principle (Artificial Recursive Creation), first proposed in *Infinite Architects* (Eastwood, 2026): that capability in intelligent systems scales super-linearly with recursive depth. The principle is expressed mathematically as U = I x R^alpha, where effective capability (U) scales with base intelligence (I) multiplied by recursive depth (R) raised to an empirically determined power alpha. Preliminary evidence across multiple AI models confirms alpha > 1, supporting the core claim that recursion is a multiplicative amplifier of intelligence rather than a mere additive accumulator.

Experiments

All experiment scripts, results, and data are in `experiments/`.

Links

- **OSF DOI:** https://doi.org/10.17605/OSF.IO/6C5XB
- **GitHub:** https://github.com/MichaelDariusEastwood/arc-principle-validation

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